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Statistical and Machine Learning Models in Credit Scoring: A Systematic Literature Survey

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Abstract

In practice, as a well-known statistical method, the logistic regression model is used to evaluate the credit-worthiness of borrowers due to its simplicity and transparency in predictions. However, in literature, sophisticated machine learning models can be found that can replace the logistic regression model. Despite the advances and applications of machine learning models in credit scoring, there are still two major issues: the incapability of some of the machine learning models to explain predictions; and the issue of imbalanced datasets. As such, there is a need for a thorough survey of recent literature in credit scoring. This article employs a systematic literature survey approach to systematically review statistical and machine learning models in credit scoring, to identify limitations in literature, to propose a guiding machine learning framework, and to point to emerging directions. This literature survey is based on 74 primary studies, such as journal and conference articles, that were published between 2010 and 2018. According to the meta-analysis of this literature survey, we found that in general, an ensemble of classifiers performs better than single classifiers. Although deep learning models have not been applied extensively in credit scoring literature, they show promising results.

Keywords: Credit Scoring, Statistical Learning, Machine Learning, Deep Learning, Systematic Literature Survey

1. Introduction

- 2 The history of credit scoring dates back to the 1950s (Thomas et al., 2002).
- 3 During those early years, credit decisions were made using a judgmental approach commonly known as 5C's approach (Thomas, 2000):
- 4 Character: do you know the person or their family?

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- 6 Capital: how much is being asked for?
- 7 Collateral: what is the borrower willing to put up from their resources?
- 8 Capacity: what is their repayment ability?
- 9 Condition: what are the conditions in the market?

10 The drawback of using the 5C's approach was the incapability of processing a
 11 huge number of applications daily. This resulted in the advent of scorecards,
 12 which make consistent nonjudgmental decisions that treat all borrowers fairly.
 13 The scorecards generate a score which quantifies the risk of lending money to
 14 borrowers.

15 When a borrower applies for a loan, the financial institution, without loosing
 16 generality hereafter referred to as the bank, collects information from the bor-
 17 rower. This information is called application data and it consists of *demographic*
 18 *information*, e.g., the number of dependents, time at current address, time at
 19 current employment, etc. The *bureau information* is also collected from the
 20 local bureaus, and it includes the number of enquiries, judgments, number of
 21 delinquencies, etc. Once the accepted population, i.e. borrowers who have been
 22 granted loans, are identified, their loan repayment history is tracked for a period
 23 of time, e.g. 24-months. A *target flag* (i.e. good/bad flag) gets created based on
 24 loan repayment history of the accepted population. If the number of days past
 25 due (or missed payments) is less than a certain number of days, e.g. 90 days,
 26 then the borrower is flagged as a *good* borrower, otherwise a *bad* borrower. The
 27 known goods and bads are then used to develop a scorecard. The cut-off score is
 28 determined by the *Kolmogorov-Smirnov statistic*, and it measures the distance
 29 between the cumulative distribution of goods and bads. The score which gives
 30 the maximum distance between the distribution of goods and bads is regarded
 31 as the cut-off score and is used to predict the goods and the bads. If a score
 32 of a borrower is larger than or equal to the cut-off score, then the borrower
 33 is predicted as a good borrower otherwise a bad borrower. The scorecard is
 34 then applied to the rejected population to predict goods and bads and this is
 35 referred to as *reject inference* (Siddiqi, 2005). The final application scorecard
 36 is built on the accepted and rejected populations, i.e. Through-The-Door (TTD)
 37 population. Please see Figure 1 for a schematic view of data flowchart for appli-
 38 cation scorecards. Traditionally, financial institutions use a logistic regression to
 39 score borrowers. The choice of using the logistic regression model is due to it's
 40 simplicity and transparency. The scores from logistic regression are calculated
 41 using the following formula (Siddiqi, 2005)

$$Score = \frac{pdo}{\ln(2)} \times \left(\beta_0 + \sum_{i=1}^n \beta_i \sum_{j=1}^{k_i} WOE_{i,j} \right), \quad (1)$$

42 where

$$WOE_{i,j} = \ln \left(\frac{F_{G_j}}{F_{B_j}} \right) \quad (2)$$

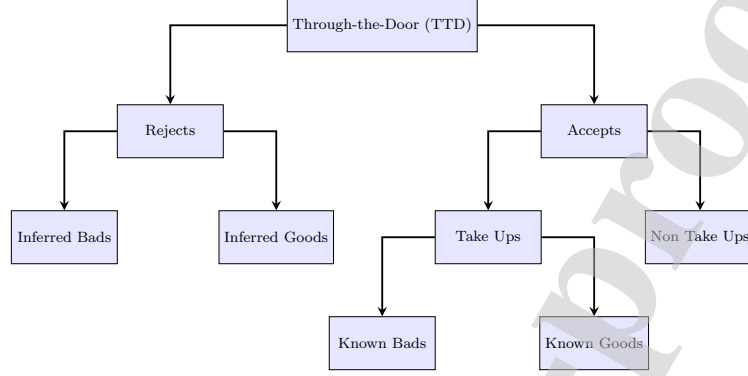


Figure 1: Schematic view of data flowchart for application scorecards.

and n is the number of features, k_i is the number of groups or attributes in the i^{th} feature, pdo is points double the odds and is used as a scaling factor and F_{G_j} and F_{B_j} are distribution of good borrowers and bad borrowers in the j^{th} attribute in feature i , respectively. The β_i coefficients are estimated using *Maximum Likelihood Estimation* (Myung, 2003).

In literature, sophisticated machine learning (ML) models can be found that can replace the logistic regression model. In spite of high accuracies from ML models; ML models are generally unable to explain their predictions. Financial institutions are regulated entities and are required to be transparent in their decisions when using application scorecards. However, techniques that can deduce rules to mitigate the lack of transparency without even compromising accuracy are suggested in the literature, e.g. (Baesens et al., 2003). In this paper, we aim to present a systematic literature survey of statistical and machine learning models which are employed in credit scoring between 2010 and 2018 and propose a guiding ML framework for credit scoring. The remainder of this paper is organised as follows. In Section 2 we cover the methodology followed for conducting this systematic literature survey. In Section 3 we highlight feature selection/engineering methods. The learning models are covered in Section 4. In Section 5 we present evaluation metrics. The data imbalance is covered in Section 6. In Section 7 we cover model transparency. In Section 8 we discuss limitations and assumptions of different models. In Section 9 we discuss emerging trends. Section 10 discusses limitations in literature. In Section 11 we discuss results. In Section 12 we propose a guiding framework for machine learning in credit scoring and finally Section 13 provides conclusion and future work.

2. Survey Methodology

This study employs a systematic literature survey approach to

- 70 1) systematically review the most commonly used statistical and machine learn-
- 71 ing techniques in credit scoring;
- 72 2) identify limitations in literature;
- 73 3) propose a guiding machine learning framework to perform credit scoring;
- 74 4) point to emerging directions.

75 In this survey, the statistical techniques considered include Linear Discrimi-

76 nant Analysis (LDA), Logistic Regression (LR) and Naïve Bayes (NB), and the

77 machine learning include k -Nearest Neighbor (k -NN), Decision Trees (DTs),

78 Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), Ran-

79 dom Forests (RFs), Boosting, Extreme Gradient Boost (XGBoost), Bagging,

80 Restricted Boltzman Machines (RBMs), Deep Multi-Layer Perceptron (DMLP),

81 Convolutional Neural Networks (CNNs) and Deep Belief Neural Networks (DBNs).

82 Note that the list of these techniques for both statistical and machine learning

83 is not exhaustive, the literature provides a myriad of techniques that are used to

84 model credit scoring. However, this study only reviews the most commonly used

85 techniques as it would be almost impossible to look at all techniques applied in

86 credit scoring.

87 A process flowchart of the systematic literature survey methodology is pre-

88 sented in Figure 2. Note that this study follows the methodology of the recently

89 published systematic literature survey on bankruptcy prediction models (Alaka

90 et al., 2018). The search words of the current study are “Statistical Learning

91 in Credit Scoring”, “Machine Learning in Credit Scoring” and “Deep Learning

92 in Credit Scoring”. The inclusion criterion for articles is based on the recently

93 published work in credit scoring and the period considered is the year 2010 to

94 the year 2018. The study selects peer-reviewed journal and conference articles

95 as these are considered to be of high quality (Schlosser, 2007). The articles are

96 selected by reading the abstract and the conclusion, in some instances the en-

97 tire article is read. This review is based on articles that are written in English

98 only. This overlooks one of the requirements in systematic literature review

99 where language constraints are discouraged. The following databases are used

100 in searching papers: Google Scholar, Science Direct, IEEEExplore, ACM and

101 SpringerLink. These databases include studies which are undertaken all over

102 the world, hence geographical bias is removed.

103 All unpublished work and dissertations are not included in this current study.

104 In the end, 74 primary studies were selected for this systematic literature survey.

105 The primary studies include models in their hybrid form (i.e. where feature se-

106 lection/feature engineering is combined with a classifier or ensemble classifiers)

107 and in their standalone form. A meta-analysis of the results from the selected

108 articles is done by producing summary tables, pie-charts and histograms. The

109 German and Australian credit datasets are the most frequently used datasets

110 in credit scoring, hence we used these two datasets for model performance com-

111 parisons.

112 2.1. Existing Literature Surveys on Bankruptcy Prediction or Credit Scoring

113 There are several literature surveys on bankruptcy prediction or credit scor-

114 ing (Bellovary et al., 2007; Abdou and Pointon, 2011; Lin et al., 2012; Wang

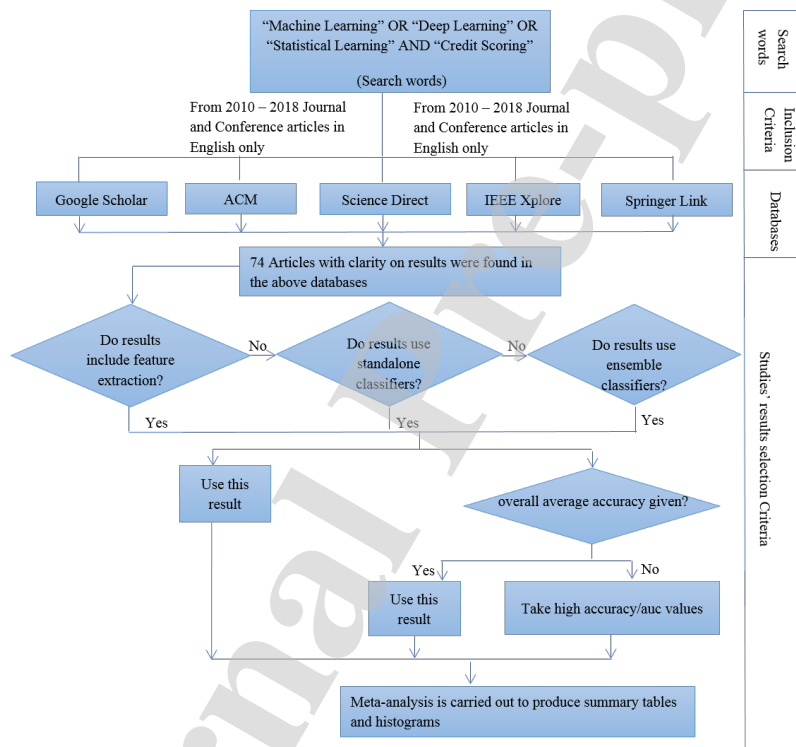


Figure 2: Process flowchart of the systematic literature survey methodology.

Survey paper	Articles searched	Years	Objective	Difference from this survey
Bellovary et al. (2007)	165	1965–2010	✓ Investigation of model type by decade ✓ Analysis of Number of features used by decade ✓ Compare model performance	✓ Transparency is not reported ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Abdou and Pointon (2011)	214	Not specified	✓ Comparing models based on performance	✓ Transparency is not reported ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Lin et al. (2012)	130	1995–2010	✓ Models are compared based on design, datasets and baselines	✓ Transparency is not covered ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Wang et al. (2015)	Not specified	Not specified	✓ Comparing models based on performance	✓ Transparency is not reported ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Louzada et al. (2016)	187	1992–2015	✓ Proposing a new method for scoring ✓ Compare traditional techniques ✓ Conceptual discussion	✓ Transparency is not reported ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Devi and Radhika (2018)	6	Not specified	✓ Compare model performance	✓ Transparency is not reported ✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported
Alaka et al. (2018)	49	2010–2015	✓ Compare models based on accuracy, transparency, data size capability, data dispersion and etc.	✓ Deep learning models are not covered ✓ Nature of datasets (balanced vs imbalanced) is not reported

Table 1: Existing literature surveys on bankruptcy prediction or credit scoring and their differences from this survey paper.

et al., 2015; Louzada et al., 2016; Devi and Radhika, 2018; Alaka et al., 2018). We briefly highlight the objectives of each literature survey and show where they differ with our study in Table 1. The literature surveys in Table 1 focus mostly on comparing models based on their performances. However, the listed literature surveys ignore key factors in credit scoring, such as model transparency and the nature of datasets. Furthermore, deep learning models are not considered in the listed literature surveys in Table 1.

Among the existing literature surveys shown in Table 1, Louzada et al. (2016) provide the most recent and the most comprehensive systematic literature survey in credit scoring. The main objective of Louzada et al. (2016) is “proposing a new method for rating, comparing traditional techniques, conceptual discussions, feature selection, literature review, performance measures studies and, at last, other issues”. Their literature survey covers a period from January 1992 to December 2015 and focuses on 12 questions on the conceptual scenario over the techniques. Comparing (Louzada et al., 2016) with our literature survey, there is an overlap from 2010 to 2015. However, our literature survey includes the most recent years. According to the best of our knowledge, there is no systematic literature survey for credit scoring that has been published which encompasses, statistical learning, traditional machine learning and deep learning models.

3. Feature Selection and Feature Engineering

This section provides a review of most commonly used *feature selection* (FS) and *feature engineering* (FE) techniques used in credit scoring. The distinction between feature selection and feature engineering is that the feature selection selects a subset of features from the entire feature set whereas feature engineering creates a new set of features from the existing features. Albeit, Liang et al. (2015) investigated the effect of feature selection and concluded that performing feature selection does not always improve the prediction performance, the studies in (Brown and Mues, 2012; Bijak and Thomas, 2012; Chen and Li, 2010; Chen and Shi, 2013; Chen and Xiang, 2017) showed that removing redundant features can improve model performance in credit scoring. There are also studies (Chi and Hsu, 2012; Back et al., 1996; Oreski and Oreski, 2014) which employed meta-heuristic approach such as *Genetic Algorithm* for feature selection. We divided the feature selection into *filter*, *wrapper* and *embedded* methods.

3.1. Filter Methods

The filter methods perform feature selection based on the univariate analysis of the features without using a predictor. They compute a score for each feature and select a subset of features based on their scores. In the following, we present most commonly used filter methods in credit scoring.

3.1.1. F-score

F-score measures the discrimination of two sets of real numbers (Chen and Li, 2010). Given m training vectors $\mathbf{x}_k$, where $k = 1, 2, \dots, m$, and if the number of positive and negative instances are $m^{(+)}$ and $m^{(-)}$, respectively, then the F-score of the i^{th} feature $F(i)$ is defined as follows

$$F(i) = \frac{(\bar{x}_i^{(+)} - \bar{x}_i)^2 + (\bar{x}_i^{(-)} - \bar{x}_i)^2}{\frac{1}{m^{(+)}-1} \sum_{k=1}^{m^{(+)}} (x_{k,i}^{(+)} - \bar{x}_i^{(+)})^2 + \frac{1}{m^{(-)}-1} \sum_{k=1}^{m^{(-)}} (x_{k,i}^{(-)} - \bar{x}_i^{(-)})^2} \quad (3)$$

where $x_{k,i}$ is the i^{th} feature of the k^{th} sample, $\bar{x}_i$, $\bar{x}_i^{(+)}$ and $\bar{x}_i^{(-)}$ are the averages of the i^{th} feature of the whole, positive, and negative datasets, respectively (Chen and Li, 2010). The numerator indicates the discrimination between the positive and negative sets, and the denominator indicates the one within each of the two sets. The larger the F-score value is, the more discriminative power the feature has (Chen and Li, 2010; Song et al., 2017).

3.1.2. Rough Set Theory

In rough set theory, the sample dataset is called information system, denoted by $I = (U, A)$, where U is a non-empty finite set of observations called the universe and A is a non-empty finite set of features. Firstly, a notion of *indiscernibility relation* on a universe set is defined. With every subset $B \subseteq A$, an indiscernibility relation on U is defined as follows

$$I(B) = \{(\mathbf{x}, \mathbf{y}) \in U \times U : f_i(\mathbf{x}) = f_i(\mathbf{y}), \forall i \in B\}, \quad (4)$$

where $f_i : U \rightarrow V_i$ is the i^{th} feature function and V_i is a set of values associated with feature i . Feature i is redundant in B if $I(B) = I(B - \{i\})$, otherwise feature i is important in B . If all of the features in B are important, then B is said to be a reduced set of features (Pawlak, 1997; Wang et al., 2010; Zhang et al., 2016).

3.2. Wrapper Methods

The wrapper methods select a subset of features based on an evaluation metric, such as accuracy, of a pre-determined predictor. Each subset of the features are scored according to their predictive power, thus, the wrapper methods are computationally expensive compared to the filter methods. However, since they select features based on the performance of the pre-determined predictor, the wrapper methods usually outperform filter methods in credit scoring. In the following, we present the most commonly used wrapper methods in credit scoring.

3.2.1. Stepwise Selection

The stepwise feature selection has three components (Siddiqi, 2005), namely, *forward feature selection*, *backward feature elimination* and a *stepwise feature selection* which is a combination of both forward feature selection and backward

feature elimination. All three components use linear regression and a p -value for feature selection. The *forward feature selection* starts by regressing one feature and if the feature is significant according to the p -value then that feature is retained. This process is repeated by adding one feature at a time until the list of features is exhausted, and the significant features are retained and insignificant features are discarded. The *backward feature elimination* works the opposite way, you start with the entire set of features and you keep discarding features with p -values less than the chosen level of significance (Tsai, 2009). The *stepwise feature selection* adds and removes features.

3.2.2. Genetic Algorithm

A genetic algorithm (Mitchell, 1996) uses genetic inspired operators to evolve an initial population into a new population. These operators are the *selection*, *crossover* and *mutation*. Each population comprises of chromosomes (a string of bits (0/1)) that represent genetically encoded individual solutions to a specific problem. Selection operator selects chromosomes in the population (a set of solutions) for reproduction. Crossover operator randomly chooses a position and exchanges the subsequent bits before and after that position between two chromosomes to create two offsprings. Mutation operator randomly flips some of the bits in a chromosome. Each individual has a fitness score assigned to them, which represents its ability of be selected. For feature selection, the individuals are subsets of features that are encoded as bits where the i^{th} feature is selected if the corresponding bit is set to 1. The fitness value is some measure of model performance, such as classification accuracy. A new population is evolved by using operators of crossover, where selection is based on individual's fitness function and its ability to reproduce the next generation (Kozeny, 2015). Hence, genetic algorithm is an evolutionary algorithm problem that is solved by searching a space (in our case a space of features). However, the key elements of problem solving by search are *exploitation* and *exploration*. The exploitation uses the information from previously visited points to determine the prospect of finding profitable regions to be visited next and the exploration is the process of visiting new regions of a search space to uncover promising offsprings (Crepinsek et al., 2013). "How and when to control and balance exploration and exploitation in the search process to obtain even better fitness results and/or convergence faster are still on-going research" Liu et al. (2009). It is key that a diversified population is maintained during the whole search process. The measure of diversity is *entropy* and it represents the amount of population disorder, where an increase in entropy results in an increase in diversity (Crepinsek et al., 2013).

3.3. Embedded Methods

The embedded methods optimize an objective function or learning classifier with a goodness-of-fit term and a penalty for a large number of features (Cadenas et al., 2013). The most commonly used embedded method in credit scoring is the least absolute shrinkage and selection operator (LASSO) (Tibshirani,

232 2011). Given a linear regression model to predict dependent variable y_i using
 233 independent variables $x_{i,j}$ s for i^{th} sample in the training dataset, i.e.

$$y_i = \sum_{j=1}^n x_{i,j} \beta_j + \beta_0, \quad (5)$$

234 where $j = 1, 2, \dots, n$, the LASSO (Tibshirani, 2011) solves the L_1 -penalized
 235 regression problem of finding the set of parameters $\beta = \{\beta_j\}_{j=1}^n$ to minimize

$$\sum_{i=1}^m \left(y_i - \sum_j x_{i,j} \beta_j \right)^2 + \lambda \sum_{j=1}^n |\beta_j|, \quad (6)$$

236 where where m is the number of instances in the training dataset and $\lambda \sum_{j=1}^n |\beta_j|$
 237 represents the penalty term. The penalty term reduces coefficients β_j and sim-
 238 plifies the linear regression model. Thus, the LASSO performs variable selection
 239 and model shrinkage based on the magnitude of β_j s.

240 3.4. Feature Engineering

241 The original features may be dependant on each other which may reduce
 242 the performance of a predictor. To tackle this problem, feature engineering is
 243 used to create a new set of engineered features from the original ones. The
 244 feature engineering methods can either decrease or increase the dimensionality
 245 of feature vectors (Zheng and Casari, 2018). In the following, we provide a brief
 246 review of most commonly used feature engineering methods in credit scoring.

247 3.4.1. Principal Component Analysis

248 Principal Component Analysis (PCA) aims to transform/compress the data
 249 from a higher dimensional space $\mathbb{R}^n$ to a lower dimensional space $\mathbb{R}^k$ (Sehgal
 250 et al., 2014), where $k \ll n$. Note that relevant feature information (variance) is
 251 not lost during this transformation. The reduced feature space consists of *prin-*
 252 *icipal components* which are orthogonal to each other. Each principal component
 253 represents the direction of maximum variance. The principal components are
 254 computed via *eigenvalue* decomposition of the *covariance matrix* of features.
 255 The resulting *eigenvectors* from the decomposition are used as principal com-
 256 ponents. The following steps are used when constructing principal components:

- 257 1. Standardize the n dimensional dataset. Each feature $x_{i,j}$ of the feature
 258 vector $\mathbf{x}_i$ is standardised according to

$$x_{i,j} \leftarrow \frac{x_{i,j} - \mu_j}{\sigma_j}, \quad (7)$$

259 where μ_j and σ_j are the mean and standard deviation of j^{th} feature.

- 260 2. Construct the *covariance matrix* Σ . The covariance matrix is computed
 261 as

$$\Sigma = \frac{1}{m-1} \sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^T, \quad (8)$$

where $\mathbf{x}_i$ is assumed a column vector.

3. Apply eigenvalue decomposition on the covariance matrix to compute eigenvalues (λ_j s) and eigenvectors ($\mathbf{e}_j$ s).
4. Select k eigenvectors that correspond to the k largest eigenvalues, where k is the dimensionality of the new feature subspace. Generally, the *variance explained ratio* is used to select the number of principal components (eigenvectors). For a set of all sorted eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n, \quad (9)$$

the variance explained ratio is defined as follows

$$\frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^n \lambda_j}. \quad (10)$$

The chosen k eigenvectors will have a high total explained variance.

5. Finally, use k eigenvectors and project each feature vector $\mathbf{x}$ to k -subspace, i.e.,

$$\hat{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu})^T [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k], \quad (11)$$

where $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T$ is the mean vector and $\hat{\mathbf{x}} \in \mathbb{R}^k$ is the projection of $\mathbf{x}$ onto k -subspace.

3.4.2. Autoencoders

Figure 3 shows an autoencoder which is a type of neural network with three layers, namely; input, hidden and output layers. For a typical autoencoder, the input variables are the same as the target variables. The autoencoder learns representations of its inputs at hidden layer and tries to reconstruct its inputs from the learned representations by optimising a cost function on a training dataset, i.e.

$$E = \frac{1}{2} \sum_{i=1}^m \|a_2(a_1(\boldsymbol{\alpha}^{(1)} \mathbf{x}_i) \boldsymbol{\alpha}^{(2)}) - \mathbf{x}_i\|_2^2, \quad (12)$$

where $\boldsymbol{\alpha}^{(1)}$, $\boldsymbol{\alpha}^{(2)}$ are weights and a_1 , a_2 are activation functions between input and hidden layer, hidden layer and output layer, respectively. Once $\boldsymbol{\alpha}^{(1)}$ and $\boldsymbol{\alpha}^{(2)}$ are learned, a feature vector $\mathbf{x}$ is mapped to $\hat{\mathbf{x}}$ as follows $\hat{\mathbf{x}} = \sigma_1(\boldsymbol{\alpha}^{(1)} \mathbf{x})$ which is further used as engineered feature vector. One can choose different number of hidden layers for the autoencoder to learn complex representations in the data. Furthermore, the autoencoder can both decrease or increase the dimensionality of the projected vectors $\hat{\mathbf{x}}$ with respect to the dimensionality of the input feature vector $\mathbf{x}$.

3.4.3. Linear Discriminant Analysis

Linear discriminant analysis (LDA) was first introduced by Fisher (1936). The following outlines the steps taken when using LDA for feature selection (Martinez and Kak, 2001).

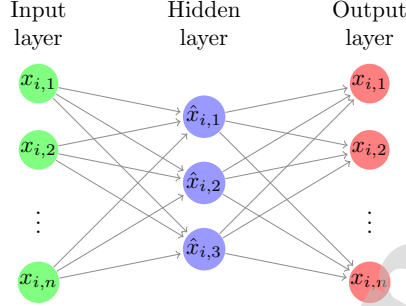


Figure 3: An example of an autoencoder with one hidden layer. The green circles represent inputs, the blue circles represent engineered features, the red circles are reconstructed inputs.

- 294 1. Calculate n -dimensional mean vectors $\boldsymbol{\mu}^{(+1)}$ and $\boldsymbol{\mu}^{(-1)}$ for “good” (or
295 positive (+1)) and “bad” (or negative (-1)) classes, respectively:

$$\boldsymbol{\mu}^{(+1)} = \frac{1}{m^{(+1)}} \sum_{k=1}^{m^{(+1)}} \mathbf{x}_k^{(+1)}, \boldsymbol{\mu}^{(-1)} = \frac{1}{m^{(-1)}} \sum_{k=1}^{m^{(-1)}} \mathbf{x}_k^{(-1)}. \quad (13)$$

- 296 2. Construct the within-class scatter matrix $\mathbf{S}_w$ and the between-class scatter
297 matrix $\mathbf{S}_b$:

$$\mathbf{S}_w = \sum_{\forall c \in \{+1, -1\}} \frac{m^{(c)}}{m^{(+1)} + m^{(-1)}} \sum_{k=1}^{m^{(c)}} \left(\mathbf{x}_k^{(c)} - \boldsymbol{\mu}^{(c)} \right) \left(\mathbf{x}_k^{(c)} - \boldsymbol{\mu}^{(c)} \right)^T \quad (14)$$

$$\mathbf{S}_b = \sum_{\forall c \in \{+1, -1\}} \frac{m^{(c)}}{m^{(+1)} + m^{(-1)}} \left(\boldsymbol{\mu}^{(c)} - \boldsymbol{\mu} \right) \left(\boldsymbol{\mu}^{(c)} - \boldsymbol{\mu} \right)^T \quad (15)$$

298 where $\boldsymbol{\mu} = \frac{m^{(+1)}}{m^{(+1)} + m^{(-1)}} \boldsymbol{\mu}^{(+1)} + \frac{m^{(-1)}}{m^{(+1)} + m^{(-1)}} \boldsymbol{\mu}^{(-1)}$ is the overall mean of
299 the dataset.

- 300 3. Apply eigenvalue decomposition on the matrix $\mathbf{inv}(\mathbf{S}_w) \mathbf{S}_b$ to compute
301 eigenvalues (λ_j s) and eigenvectors ($\mathbf{e}_j$ s), where $\mathbf{inv}(\cdot)$ is matrix inverse.
302 4. Similar to PCA, select k eigenvectors that correspond to the k largest
303 eigenvalues.
304 5. Finally, use k eigenvectors and project each feature vector $\mathbf{x}$ to k -subspace,
305 i.e.,

$$\hat{\mathbf{x}} = \mathbf{x}^T [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k], \quad (16)$$

306 where $\hat{\mathbf{x}} \in \mathbb{R}^k$ is the projection of $\mathbf{x}$ onto k -subspace.

307 4. Supervised Learning

308 Credit scoring is a *supervised learning* problem, specifically it is a binary
309 classification problem, where the aim is to classify good borrowers and bad

310 borrowers.. In this section we first define supervised learning problem and then
 311 present most commonly used statistical and machine learning methods in credit
 312 scoring.

313 4.1. Supervised Learning Problem

314 In the context of credit scoring, supervised learning problem can be defined
 315 as searching for a model $\phi : \mathbb{R}^n \mapsto \{+1, -1\}$ which maps a feature vector $\mathbf{x} \in \mathbb{R}^n$
 316 to a predicted class label $\hat{y} \in \{+1, -1\}$, i.e

$$\phi_{\alpha} : \mathbf{x} \rightarrow \hat{y}, \quad (17)$$

317 α is the set of the model parameters. For the ease of reading ϕ_{α} and ϕ are used
 318 interchangeably.

In order to learn the model parameters, information from the previously
 processed applicants are collected to form a labeled dataset

$$\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^m,$$

319 where $\mathbf{x}_k \in \mathbb{R}^n$ denotes k^{th} applicant's feature vector and $y_k \in \{+1, -1\}$ is
 320 the corresponding label from the set of classes (good borrower = +1) and (bad
 321 borrower = -1). The dataset $\mathcal{D}$ is split into a training set $\mathcal{D}_{train} \subset \mathcal{D}$ and a test
 322 dataset $\mathcal{D}_{test} \subset \mathcal{D}$ where $\mathcal{D}_{train} \cap \mathcal{D}_{test} = \emptyset$ and $\mathcal{D}_{train} \cup \mathcal{D}_{test} = \mathcal{D}$.

323 Once the training and testing datasets are formed, the parameters α of the
 324 model ϕ are learned on the training dataset $\mathcal{D}_{train}$ by minimising a cost function
 325 C

$$C(\mathcal{D}_{train}, \phi) = \sum_{\forall \mathbf{x}_k \in \mathcal{D}_{train}} d(y_k, \phi(\mathbf{x}_k)), \quad (18)$$

326 $d(y_k, \phi(\mathbf{x}_k))$ is a distance measure, such as the Mean Squared Error (MSE) or
 327 Cross-Entropy, between y_k and its prediction $\hat{y}_k = \phi(\mathbf{x}_k)$ from the model. The
 328 optimal parameters α^* of the model ϕ are obtained according to

$$\alpha^* = \underset{\alpha}{\operatorname{argmin}} C(\mathcal{D}_{train}, \phi_{\alpha}). \quad (19)$$

329 The performance of the learned model ϕ_{α^*} is assessed on the test dataset $\mathcal{D}_{test}$.

330 4.2. Statistical Learning Models

331 This section discusses popular statistical learning techniques in credit scor-
 332 ing.

333 4.2.1. Linear Discriminant Analysis

334 The *Linear Discriminant Analysis (LDA)* develops a linear combination of
 335 independent features which yield the largest mean difference between classes.
 336 Under the assumption of equal class covariances Rao (1948), we have the follow-
 337 ing solution in equation (20) for a binary classification problem, such as credit

scoring. For a given feature vector $\mathbf{x}$ we decide for class $\hat{y} = +1$ if the expression on the right hand side is greater than the expression on the left hand side, otherwise we decide for class $\hat{y} = -1$

$$(\boldsymbol{\mu}^{(+)} - \boldsymbol{\mu}^{(-)})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \geq \log(P(y = -1)) - \log(P(y = +1)), \quad (20)$$

where $P(\cdot)$ denotes probability.

4.2.2. Logistic Regression

The Logistic Regression (LR) is the most commonly used statistical model in credit scoring due to interpretability of its decisions. The LR model is defined as

$$P(y = +1|\mathbf{x}) = \frac{1}{1 + \exp(\alpha_0 + \boldsymbol{\alpha}^T \mathbf{x})} \quad (21)$$

and

$$P(y = -1|\mathbf{x}) = 1 - P(y = +1|\mathbf{x}) = \frac{\exp(\alpha_0 + \boldsymbol{\alpha}^T \mathbf{x})}{1 + \exp(\alpha_0 + \boldsymbol{\alpha}^T \mathbf{x})}, \quad (22)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the feature vector, $P(y = +1|\mathbf{x})$ is the probability of classifying $\mathbf{x}$ as a good borrower, $P(y = -1|\mathbf{x})$ is the probability of classifying $\mathbf{x}$ as a bad borrower and $\{\alpha_0, \boldsymbol{\alpha}\}$ are the model parameters estimated by using, e.g., maximum likelihood estimation on the training dataset (Myung, 2003). Once the model parameters are estimated, the decision on an input feature vector $\mathbf{x}$ is made in favor of $\hat{y} = +1$ if

$$P(y = +1|\mathbf{x}) \geq P(y = -1|\mathbf{x}), \quad (23)$$

which is equivalent to the following decision rule

$$\hat{y} = \begin{cases} +1 & \text{for } 1 \geq \exp(\alpha_0 + \boldsymbol{\alpha}^T \mathbf{x}); \\ -1 & \text{otherwise.} \end{cases} \quad (24)$$

4.2.3. Naïve Bayes

The Naïve Bayes (NB) classifier is based on Bayes decision rule Duda et al. (2001). The adjective "naïve" comes from the assumption that the features in a dataset are mutually independent. The Naïve Bayes classifier decides for class $y = +1$ over $y = -1$ if

$$P(y = +1|\mathbf{x}) \geq P(y = -1|\mathbf{x}), \quad (25)$$

where

$$P(y = +1|\mathbf{x}) = \frac{p(\mathbf{x}|y = +1)P(y = +1)}{p(\mathbf{x})} \quad (26)$$

and

$$P(y = -1|\mathbf{x}) = \frac{p(\mathbf{x}|y = -1)P(y = -1)}{p(\mathbf{x})}. \quad (27)$$

361 The probability density $p(\mathbf{x})$ of observing $\mathbf{x}$ is defined according to

$$p(\mathbf{x}) = p(\mathbf{x}|y = +1)P(y = +1) + p(\mathbf{x}|y = -1)P(y = -1) \quad (28)$$

362 and the conditional density functions $p(\mathbf{x}|y = -1)$ and $p(\mathbf{x}|y = +1)$ are modelled
363 according to

$$p(\mathbf{x}|y = -1) = \frac{1}{(2\pi)^{\frac{m}{2}} |\Sigma^{(-)}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}^{(-)})^T \text{inv}(\Sigma^{(-)}) (\mathbf{x} - \boldsymbol{\mu}^{(-)})\right\} \quad (29)$$

364 and

$$p(\mathbf{x}|y = +1) = \frac{1}{(2\pi)^{\frac{m}{2}} |\Sigma^{(+)}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}^{(+)})^T \text{inv}(\Sigma^{(+)}) (\mathbf{x} - \boldsymbol{\mu}^{(+)})\right\}, \quad (30)$$

365 where $\Sigma^{(-)}$ and $\Sigma^{(+)}$ are covariance matrixes computed on negative and positive
366 instances from the training dataset. For the Naïve Bayes classifier, $\Sigma^{(-)}$ and
367 $\Sigma^{(+)}$ are diagonal matrices.

368 Note that $p(\mathbf{x}|y = -1)$ and $p(\mathbf{x}|y = +1)$ are uni-modal multivariate Gaus-
369 sian distribution density functions which may not adequately model multi-
370 modal data. In this case, one can employ multi-variate *Gaussian Mixture Model*
371 (Reynolds, 2015) and Expectation Maximization (EM) (Dempster et al., 1977)
372 to learn the model parameters on the training dataset.

373 4.3. Machine Learning Models

374 This section discusses popular machine learning techniques in credit scoring.

375 4.3.1. *k*-Nearest Neighbor

376 A *k*-Nearest Neighbor (*k*-NN) (Henley and Hand, 1996) assigns to an input
377 feature vector $\mathbf{x}$ the class of the majority of its *k* nearest neighbors in the training
378 dataset. The nearest neighbors are determined by calculating the *Euclidean*
379 *distance* or *Mahalanobis distance* between the input feature vector $\mathbf{x}$ and the
380 training dataset $\{\mathbf{x}_k\}_{k=1}^m$. Thus the class for the new data point is

$$y = \underset{y \in \{+1, -1\}}{\text{majority}} \left[\underset{k}{\text{argmin}} \|\{\mathbf{x}_k\}_{k=1}^m - \mathbf{x}\| \right]. \quad (31)$$

381 4.3.2. Decision Trees

382 As shown in Figure 4 a Decision Tree (DT) (Duda et al., 2001) asks a series
383 of questions in order to arrive to an answer (which is a class label). The root
384 of the Decision Tree is called the *root node* and it is the most discriminating
385 feature. The *leaf nodes* denote the classes.

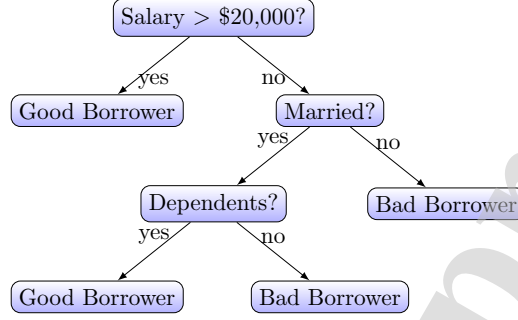


Figure 4: An example of a decision tree in credit scoring.

4.3.3. Support Vector Machines

A Support Vector Machine (SVM) (Cortes and Vapnik, 1995) uses an idea of a hyperplane (which is a decision boundary) that separates classes in a high dimensional feature space. The linear SVM focuses on maximizing the margin $\|\alpha\| = \sqrt{\sum_{i=1}^m \alpha_i^2}$ between the negative and positive hyperplanes. The correct class is assigned by using the following equation:

$$y = \begin{cases} +1, & \text{if } b + \alpha^T \mathbf{x} \geq +1 \\ -1, & \text{if } b + \alpha^T \mathbf{x} \leq -1. \end{cases} \quad (32)$$

where b is the bias. For non-linear cases, a kernel trick (Schölkopf, 2000) is used to project features into a high dimensional space.

4.3.4. Artificial Neural Networks

An Artificial Neural Network (ANN) (Mitchell, 1997) is a system which is motivated by a biological neural network system. An ANN emulates the way in which a biological neural network of the brain processes information by means of interconnected neurons (Thomas, 2000; Barboza et al., 2017; Tsai et al., 2014). Typically, a neural network consists of three layers, namely an input, hidden and output layers Tsai and Wu (2008). Essentially, training a neural network involves the process of finding optimal weights that map the input and output layers by means of *back-propagation*.

For a given input feature vector $\mathbf{x}$, a three-layer ANN computes the output $\hat{\mathbf{y}}$ according to

$$\hat{\mathbf{y}} = a_2(a_1(\alpha^{(1)}\mathbf{x} + \alpha_0^{(1)})\alpha^{(2)} + \alpha_0^{(2)}), \quad (33)$$

where $(\alpha_0^{(1)}, \alpha^{(1)})$, $(\alpha_0^{(2)}, \alpha^{(2)})$ are weights and a_1, a_2 are activation functions between input and hidden layer, hidden layer and output layer, respectively. The parameters are learned on a training set. The ANN performs final decision by applying a decision function, such as soft-max, on $\hat{\mathbf{y}}$. The ANN was first applied in credit scoring by Odom and Sharda (1990).

410 4.3.5. Random Forests

411 A Random Forest (RF) is an ensemble of decision trees (Breiman, 2001), i.e.
 412 K decision trees are built on bootstrapped samples with m observations. Each
 413 decision tree is developed using a subset of randomly chosen k features. Each
 414 decision tree will give a class of a new feature vector. Thereafter, for overall
 415 classification, a RF assigns the class of the new feature vector by using majority
 416 vote based on the outputs from the decision trees.

417 4.3.6. Boosting

418 The Boosting works by estimating multiple models iteratively and assigning
 419 weights to data instances (Freund and Schapire, 1997). Boosting starts by
 420 developing a weak model (i.e. shallow decision tree). Thereafter a better model
 421 that will address errors of the previous model is developed. The instances which
 422 were incorrectly classified by the previous model will be assigned higher weights.
 423 The popular boosting technique is *Adaptive Boosting* (Ada-boost). The Ada-
 424 boost assigns a class to an input feature vector $\mathbf{x}$ in the following way

$$\hat{y} = \text{sign} \left(\sum_{t=1}^T \alpha_t \phi_t(\mathbf{x}) \right), \quad (34)$$

425 where α_t is the weight of classifier $\phi_t(\mathbf{x})$ and T is the total number of classifiers.
 426 The parameters α_t s are learned on the training dataset.

427 4.3.7. Extreme Gradient Boost

428 The XGBoost (Chen and Guestrin, 2016) is short for extreme gradient boost-
 429 ing. The XGBoost is famously known for its processing speed and performance.
 430 It works similar to gradient boosting but builds the decision trees in parallel
 431 instead of building the decision trees in a series (Nobre and Neves, 2019a). The
 432 optimization function to minimize is

$$\mathcal{L}^{(t)} = \sum_{k=1}^n l(y_k, \hat{y}_k^{(t-1)} + \phi_t(\mathbf{x}_k)) + \Omega(\phi_t) \quad (35)$$

433 where $l(\cdot)$ is a loss function and $\Omega(\phi_t)$ is a regularization term that penalizes
 434 complexity of the model. The goal of XGBoost is to find the ϕ_t which minimizes
 435 the objective function $\mathcal{L}^{(t)}$ (Xia et al., 2017).

436 4.3.8. Bagging

437 The Bagging classifier is also known as the *bootstrap aggregation* (Breiman,
 438 1996). The Bagging technique takes K bootstraps from the underlying datasets
 439 and builds a classifier for every bootstrap. Then a class label is assigned by
 440 using a majority vote from votes of the K classifiers

$$y = \underset{y \in \{+1, -1\}}{\operatorname{argmax}} \sum_{i=1}^K 1(y = \phi_i(\mathbf{x})), \quad (36)$$

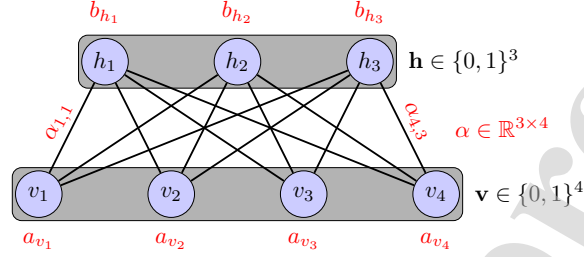


Figure 5: A restricted boltzman machine architecture with a visible layer $\mathbf{v}$ and a hidden layer $\mathbf{h}$.

where

$$1(y = \phi_i(\mathbf{x})) = \begin{cases} 1, & \text{if } y = \phi_i(\mathbf{x}); \\ 0, & \text{if } y \neq \phi_i(\mathbf{x}). \end{cases} \quad (37)$$

4.4. Deep Learning Models

Deep learning algorithms have been successfully applied in literature since the 1980s in an attempt to improve classification accuracy. Moreover, deep learning models with optimal hidden layers have been developed to reveal information not easily detectable with traditional statistical and machine learning models. The following subsections highlight the deep learning models that are used in credit risk datasets.

4.4.1. Restricted Boltzman Machines

As shown in Figure 5, a Restricted Boltzman Machine (RBM) can be perceived as an undirected neural network with two layers. The two layers can be called the hidden and the visible layers. The hidden layer is used as a feature detector while the visible layer is used to train the input data (Yu et al., 2018). Given n visible layers $\mathbf{v}$, and m hidden layers $\mathbf{h}$, the energy function is

$$E(v, h) = - \sum_{i=1}^n a_i v_i - \sum_{j=1}^m b_j h_j - \sum_{i=1}^n \sum_{j=1}^m \alpha_{ij} v_i h_j, \quad (38)$$

where a_i and b_j are biases for binary variables v_i and h_j respectively, while α_{ij} are the weights between unit i and j .

4.4.2. Deep Belief Neural Networks

Deep Belief Networks (DBNs), introduced by Hinton et al. (2006), is a class of deep neural networks. As shown in Figure 6 a typical DBN is composed of several hidden layers of RBM. Essentially, an output of a lower level RBM can be perceived as input of the higher level RBM. Figure 6 shows a graphical view of the DBN layers.

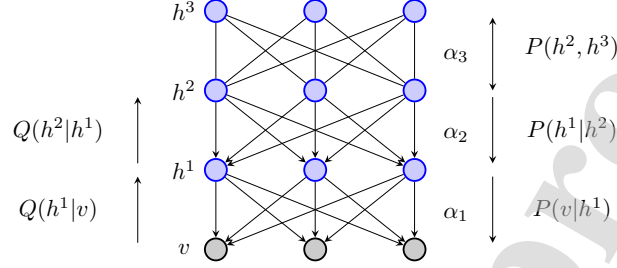


Figure 6: A deep belief neural network architecture which is composed of several hidden layers of restricted boltzman machines.

4.4.3. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) were first introduced by LeCun et al. (1989) and have been mainly used in image processing (Seo and shik Shin (2019); Lecun and Bengio (1995); Ting et al. (2019)), and on time series data (Lecun and Bengio (1995); Sezer and Ozbayoglu (2018); Zhao et al. (2017)). The convolutional neural network consists of an *input layer*, *convolutional layers*, *pooling layers* and *fully connected layers* (see Figure 7). The convolutional and pooling layers are responsible for extracting data representations Chollet (2017).

Input. A convolutional neural network takes inputs as *tensors* of shape (*height*, *width*, *channel*). In image classification, the height and width values represent the height and width of an image. The channel represents the depth/color of an image (e.g. 1 represents a gray-scale image and 3 represents an image with color). A tensor is a multidimensional array that contains numerical values or in some instances non-numeric values. The input shape is normally changed into a shape that the convolutional neural network anticipates and the input is scaled so that all values are in the $[0, 1]$ interval (Chollet, 2017). Since credit scoring data is not an image data, a 1D convolutional neural network is normally used for non-image data with the exception of speech/voice data. Hence, for credit scoring, the CNN architecture consists of an input layer which is a tensor of shape $(m, 1, n)$, where m is the number of instances and n is the number of features. Each instance is a feature vector with three channels that has a shape $(1, 1, n)$. All inputs are scaled into $[0, 1]$ interval.

Convolutional Layer. The word “convolution” refers to a mathematical operation which is a specialized kind of linear operation (Goodfellow et al., 2016). Below is a convolution function

$$(f * g)(t) = \int_{-\infty}^{\infty} f(x)g(t - x)dx. \quad (39)$$

488 A convolutional layer learns local patterns (Chollet, 2017), for example, in
 489 image classification, a convolutional layer breaks down an image into edges
 490 and textures. These local patterns are *feature maps* and are also known as
 491 *activation maps*. A convolution operation is applied on the input tensor and
 492 the *feature detector*. The feature detector also known as the *filter* or *kernel*
 493 is a tensor of shape (*height*, *width*). The feature detector slides along different
 494 locations of an input tensor to form a feature map. The feature map is a reduced
 495 and transformed input tensor. A non-linear activation function such as *ReLU*
 496 (Rectified Linear Unit) or a *Sigmoid* function is applied on activation maps to
 497 introduce non-linearity.

498 *Pooling Layer*. A pooling layer is responsible for *downsampling* feature maps.
 499 At pooling layer, either a *max pooling* or an *average pooling* is used. For max
 500 pooling, each local input location is transformed by taking the maximum value
 501 of each channel over the location, whereas for average pooling, an average value
 502 of each channel is used. A convolutional neural network has a property called
 503 *spatial invariance*, which occurs at a pooling layer. The spatial invariance as-
 504 sures that the network is not influenced by input distortions or variations. Thus,
 505 the pooling layer is capable of preserving important features.

506 *Flattening*. The pooled feature maps are then converted into a single vector
 507 by a process called *flattening*. This single vector becomes an input to a fully
 508 connected artificial neural network.

509 *Fully Connected Layer*. A Fully Connected Layer is a feed forward artificial
 510 neural network. It consists of an input layer, hidden layer/s and an output
 511 layer. At output layer, a *loss function* or *cost function* is defined as follows

$$C(y, \hat{y}) = \sum_{\forall i} d(y_i, \hat{y}_i), \quad (40)$$

512 where $y_i, \hat{y}_i \in \{+1, -1\}$ and $d(y_i, \hat{y}_i)$ is a measure between y_i and $\hat{y}_i^{(i)}$ such as
 513 the Mean Squared Error(MSE) or Cross-Entropy. The mean squared error is

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2, \quad (41)$$

514 and the cross-entropy is

$$H(y, \hat{y}) = - \sum_{i=0}^1 y_i \log \hat{y}_i. \quad (42)$$

515 A neural network tries to minimize the cost function by learning and updating
 516 the connection weights using *back propagation* (Bishop, 1995).

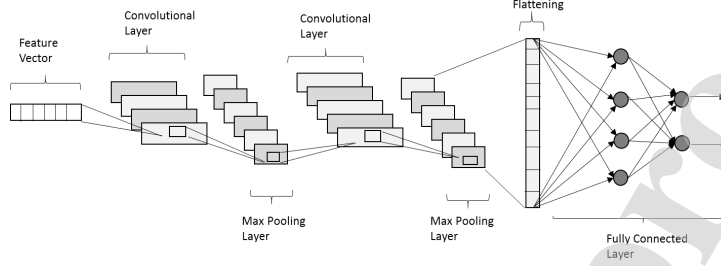


Figure 7: A 1D convolutional neural network architecture which consists of two convolutional layers, two pooling layers and a fully connected layer.

4.4.4. Deep Multi-Layer Perceptron

A Multi-Layer Perceptron with multiple hidden layers is called Deep Multi-Layer Perceptron. Deep Multi-Layer Perceptron is a directed neural network and training is performed by back-propagation. The cost or loss function for Deep Multi-Layer Perceptron uses *Softmax* and *Cross-Entropy* to update the weights. The Softmax function is

$$f_j(z) = \frac{e^{z_j}}{\sum_k e^{z_k}} \quad (43)$$

and $f_j(z) \in [0, 1]$ and z_j is a probability output from the network.

5. Evaluation Metrics

There are many evaluation metrics which are used in literature and the following metrics are the most popular metrics for assessing the performance of the models in credit scoring. The *Percentage Correctly Classified* (PCC), *Type I Error*, *Type II Error*, *Kolmogorov-Smirnov Statistic* (K-S), *Sensitivity/Recall*, *Specificity*, *Geometric-Mean* (G-mean), *F-measure* and *Area Under Receiver Operating Characteristics Curve* (AUC). Each of these metrics is shown in the sequel.

A *confusion matrix* (see Table 2) consists of *True Positives* (TP), *True Negatives* (TN), *False Positives* (FP) and *False Negatives* (FN) and is used for calculating the metrics which are discussed in this section. Based on credit scoring classification, TN is the number of borrowers who are correctly classified as non-defaults, FP is the number of non-defaulted borrowers who are incorrectly classified as defaults, FN is the number of defaulted borrowers who are incorrectly classified as non-defaults and TP is the number of borrowers who are correctly classified as defaults.

From the confusion matrix the performance metrics can be derived, such as

$$\text{PCC} = \frac{(TP + TN)}{(TP + FP + FN + TN)}, \quad (44)$$

Actual	Predicted	
	Positives	Negatives
	Positives	Negatives
	TP	FN
Negatives	FP	TN

Table 2: A confusion matrix with correct predictions, True Positives (TP) and True Negatives (TN), and incorrect predictions, False Positives (FP) and False Negatives (FN).

$$\text{Type I} = \frac{(FP)}{(FP + TN)}, \quad (45)$$

$$\text{Type II} = \frac{(FN)}{(TP + FN)}, \quad (46)$$

$$\text{Sensitivity/Recall} = \frac{TP}{(TP + FN)}, \quad (47)$$

$$\text{Specificity} = \frac{TN}{(FP + TN)}, \quad (48)$$

$$\text{G-mean} = \sqrt{\frac{TN}{(FP + TN)} \times \frac{TP}{(TP + FN)}}, \quad (49)$$

$$\text{F-measure} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Precision} + \text{Recall}}. \quad (50)$$

541 The Kolmogorov-Smirnov Statistic

$$\text{K-S} = \max |P(s|G) - P(s|B)| \quad (51)$$

542 calculates the maximum distance between the cumulative score distribution of
543 bads $P(s|B)$ and goods $P(s|G)$

$$P(s|B) = \sum_{t \leq s} p(t|B) \quad (52)$$

544 and

$$P(s|G) = \sum_{t \leq s} p(t|G) \quad (53)$$

545 respectively and where $s \in \mathbb{Z}^+$ denotes a score in $500 \leq s \leq 1000$.

546 The AUC (Tomczak and Zieba, 2015) is

$$\text{AUC} = \frac{1}{2} \left(1 + \frac{TP}{TP + FN} - \frac{FP}{FP + TN} \right) \quad (54)$$

547 and measures the discriminative power of a model between classes.

548 6. Data Imbalance

549 Imbalanced datasets occur as the number of observations in one class (re-
 550 ferred to as a minority class) in a dataset is usually much lower than the number
 551 of observations in the other class (referred to as a majority class). There are
 552 studies that have dealt with imbalanced datasets. For example, Brown and
 553 Mues (2012) showed that the random forest and gradient boosting classifiers
 554 perform very well in a credit scoring context and are able to cope comparatively
 555 well with pronounced class imbalances in the datasets. On the other hand, when
 556 faced with a large class imbalance, the C4.5 decision tree algorithm, quadratic
 557 discriminant analysis and k -nearest neighbors perform significantly worse than
 558 the best performing classifiers. Douzas and Bacao (2017) tackle the problem
 559 of imbalanced datasets by using a novel oversampling method, Self-Organizing
 560 Map-based Oversampling (SOMO).

561 There are a number of over-sampling (applied on minority class) or under-
 562 sampling techniques (applied on majority class) that can be found in literature.
 563 For example, Chawla et al. (2002) proposed Synthetic Minority Over-sampling
 564 Technique (SMOTE). The SMOTE over-samples the minority class by taking
 565 each minority class sample and creating synthetic examples (along the line seg-
 566 ments joining any/all of the k minority class nearest neighbors). Thereafter,
 567 neighbors from the k nearest neighbors are randomly chosen, depending on the
 568 amount of over-sampling required. For instance, if the amount of over-sampling
 569 needed is 300%, only three neighbors are chosen and one sample is generated in
 570 the direction of each. Synthetic samples are generated by taking the difference
 571 between the feature vector (sample) under consideration and its nearest neigh-
 572 bor. Thereafter this difference is multiplied by a random number between 0 and
 573 1, and is added to the feature vector under consideration to form a synthetic
 574 feature vector.

575 There are other techniques that neither do over-sampling nor under-sampling
 576 to deal with class imbalance, such as *wavelet data transformation* and *linear de-
 577 pendence approach*. Saia et al. (2018) proposed a discrete wavelet transformation
 578 to deal with imbalanced data in credit scoring. Wavelets are small waves and
 579 wavelet transform captures both the time and frequency domains (Nobre and
 580 Neves, 2019b). The discrete wavelet transform disintegrates a signal wave into
 581 a set of wavelets that is mutually orthogonal (Nobre and Neves, 2019b). Saia
 582 et al. (2018) approach outperformed the random forest model regardless of data
 583 distributions. Saia and Carta (2016) proposed a linear dependence approach
 584 for imbalanced data. The idea was to exploit one class (the majority class) to
 585 overcome imbalanced class distribution issue. The linear dependence is deter-
 586 mined by calculating the determinant of a square and non-square matrices. The
 587 determinant is a real number that is calculated from a matrix. When vectors
 588 are dependent, their determinant is zero. Saia and Carta (2016) used an average
 589 of sub-matrix determinants and reliability band of the majority class to classify
 590 new instances. The proposed approach performed very similarly to the random
 591 forest model.

592 7. Model Transparency

593 This section discusses different model transparency techniques. The aim of
594 model transparency is to make non-transparent models explainable. In a case
595 of credit scoring, this will help in explaining why a borrower was not granted a
596 loan. In the following, we present the most commonly used model transparency
597 techniques in credit scoring.

598 7.1. *NeuroRule*

599 The NeuroRule was first introduced by Setiono and Liu (1996). The NeuroRule
600 uses three-layer feed-forward neural network. The NeuroRule consists of
601 six steps:

- 602 Step-1) Build and train a neural network;
- 603 Step-2) Prune the neural network to remove irrelevant connections;
- 604 Step-3) Discretize the hidden unit activation values of the pruned neural network by clustering;
- 605
- 606 Step-4) Extract rules that describe the network outputs in terms of the discretized hidden activation values;
- 607
- 608 Step-5) Extract rules that describe the discretized hidden unit activation values in terms of the network inputs;
- 609
- 610 Step-6) Combine the two sets of rules extracted in steps 4 and 5 to obtain a final set of rules that relate the inputs and outputs of the network.
- 611

612 7.2. *Trepan*

613 Trepan was first introduced by Craven and Shavlik (1995). Trepan is an
614 hybrid intelligent system. It [Trepan] induces a tree to approximate any classifier's predictions. Hence, Trepan is not restricted to neural networks. Trepan
615 learns queries as opposed to normal decision trees which learn from data. At
616 each node, Trepan stores (i) a subset of training observations, (ii) a set of *query instances* and (iii) a set of constraints (the conditions that observations must
617 satisfy in order to reach the node).
618
619

620 7.3. *LIME*

621 LIME stands for Local Interpretable Model-Agnostic Explanations. LIME is
622 capable of explaining a prediction of any classifier by learning an interpretable
623 model (e.g. a Decision Tree or a linear model) around a prediction (Ribeiro
624 et al., 2016). The logic behind explaining predictions is for human subjects to
625 have trust in the predictions if actions need to be taken. For example, if a doctor
626 depends on a model to predict presence/absence of any disease, he/she will need
627 to trust the model predictions in order for him/her to prescribe medication for
628 patients. The doctor has prior knowledge in the field of medicine, he/she will
629 either accept or reject the explanation based on his/her expertise. Hence, human
630 prior knowledge plays an important role for accepting or rejecting explanations
631 for predictions. LIME is designed to provide *trust* to human subjects.

7.3.1. Interpretable Data Representation

Before acquiring explanations, a data set needs to be in a format that is understandable to humans. This is applicable, e.g., in image classification or text classification, where features in the input space need to be transformed to a vector of ones and zeroes to indicate the presence or absence of feature components. An observation $\mathbf{x} \in \mathbb{R}^n$ is transformed into $\hat{\mathbf{x}} \in \{0, 1\}^n$ for interpretability.

7.3.2. Local Fidelity

The behaviour of a classifier in the vicinity of an individual prediction determines the local faithfulness of an explanation. Let an explanation be denoted by $e \in E$ where E is a space of interpretable models, e.g., decision trees or linear models. The complexity of an explanation is given by $\Omega(e)$, e.g., for decision trees the complexity is the depth of a tree. Let $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a classifier which needs to be explained and let $\pi_{\mathbf{x}}(\mathbf{y})$ be a proximity measure between observation $\mathbf{y}$ to observation $\mathbf{x}$ which defines the locality around $\mathbf{x}$. The measure of how unfaithful is e in explaining $\phi(\mathbf{x})$ in the locality given by $\pi_{\mathbf{x}}(\mathbf{y})$ is

$$\psi\{\phi, e, \pi_{\mathbf{x}}\}, \quad (55)$$

where

$$\pi_{\mathbf{x}}(\mathbf{y}) = \exp(-D(\mathbf{x}, \mathbf{y})^2/\sigma^2) \quad (56)$$

is an exponential kernel defined on some distance D (e.g. Mahalanobis distance).

To obtain interpretability and local fidelity, an optimisation is performed to minimize equation Equation (55) and $\Omega(e)$ is kept low. Hence, the prediction explanation for observation $\mathbf{x}$ is

$$\zeta(x) = \underset{e \in E}{\operatorname{argmin}} \psi\{\phi, e, \pi_{\mathbf{x}}\} + \Omega(e). \quad (57)$$

7.3.3. Model Agnostic

For explanations to be model-agnostic, we do not need to make assumptions about ϕ . This will ensure that any prediction of *any* black-box model can be explained.

7.3.4. Sampling

The Equation (55) is approximated by drawing samples which are weighted by $\pi_{\mathbf{x}}$ around an observation of interest $\mathbf{x}$. The sampling is done by drawing non-zero elements of $\mathbf{x}$ uniformly at random (Ribeiro et al., 2016). Then $\hat{\mathbf{x}} \in \{0, 1\}^n$ is a perturbed sample which contains non-zero elements of $\mathbf{x}$. The perturbed data set is denoted by $\mathcal{R}$. The outcome/label prediction for $\hat{\mathbf{x}}$ is given by $\phi(\hat{\mathbf{x}})$.

662 7.3.5. Sparse Linear Explanations

663 This is a part where predictions are explained. In this stage an *explainer* is
664 a linear model. Hence, $e(\hat{\mathbf{x}})$ is a linear model

$$e(\hat{\mathbf{x}}) = \boldsymbol{\alpha}^T \hat{\mathbf{x}}. \quad (58)$$

665 Let $\mathbf{x}$ be the original representation of $\hat{\mathbf{x}}$. Then the Equation (55) becomes a
666 locally weighted square loss

$$\psi\{\phi, e, \pi_{\mathbf{x}}\} = \sum_{\mathbf{x}, \hat{\mathbf{x}} \in \mathcal{R}} \pi_{\mathbf{x}} (\Phi(\mathbf{x}) - e(\hat{\mathbf{x}}))^2. \quad (59)$$

667 8. Model Limitations and Imposed Assumptions

668 The Linear Discriminant Analysis (LDA) technique is one of the two popular
669 statistical techniques applied in credit scoring, and is subject to a parametric
670 assumption which is defined as the underlying multivariate normal distribu-
671 tion of the features. Eisenbeis (1978) argues that categorical features violate
672 this parametric assumption. The other model which uses two strong statis-
673 tical assumptions (i.e. i. the assumption that the features are conditionally
674 independent ii. the values of numeric features are normally distributed) is the
675 Naïve Bayes Classifier. The normally distributed assumption in Naïve Bayes
676 classifier does not always hold in other domains, hence a need to estimate other
677 continuous distributions is required (John and Langley, 1995).

678 The most popular statistical technique in credit scoring is Logistic Regression
679 (LR). The Logistic Regression assumes a linear relationship between the inputs
680 and the log odds. However, the linearity assumption does not always hold, there
681 are other cases where the relationship between the independent variables and
682 the log odds is non-linear. In such cases kernel Support Vector Machines (SVM)
683 serves as one of the non-linear classification techniques to be used. However, Yu
684 et al. (2011a) argues that the performance of SVM model is sensitive not only to
685 the algorithm for solving the Quadratic Programming problem but also to the
686 parameters setting in learning (i.e. regularized parameter balancing the classifi-
687 cation margin and tolerable misclassification errors, and the kernel parameter).
688 Yu et al. (2011a) propose Least Squares SVM (LSSVM) to solve the quadratic
689 programming problem and the design of the experiment for parameter selection
690 in SVM modeling.

691 Henley and Hand (1996) propose the use of a non-parametric technique,
692 k -nearest neighbor method in credit scoring. However, the k -nearest neighbor
693 method is costly because it requires more computations (i.e. it calculates a
694 metric distance for each data record stored during classification). Jiang (2009)
695 proposes the use of a decision tree (C4.5) in conjunction with an approach called
696 Simulating Annealing Algorithm (SAA) which performs global optimization. In
697 this study, Jiang highlights the shortcoming of the Decision Tree approach which
698 is the inability to perform global optimization because of its local search strategy,

hence the need to use SAA. Albeit the ANNs better performance when compared to other techniques in terms of accuracy, it lacks interpretability (Baesens et al., 2003). Baesens et al. (2003) propose Decompositional and Pedagogical approaches to extract rules (without compromising the accuracy) from the ANN for interpretability.

Deep Learning models have not been applied extensively in credit scoring. However, the problem with deep learning models is interpretability. Since banks are governed by regulators, banks are required to be transparent in their credit scoring process. A bank needs to tell a borrower why his/her loan application has been rejected.

9. Emerging Trends

Most banks have developed an interest in applying machine learning models (including deep learning models) in their credit scoring systems. However, the regulators still maintain that the models which are used for credit scoring should be transparent. Despite their non-transparency, machine learning models are gaining traction in credit scoring. One way of mitigating the non-transparency issue of machine learning models is to rationalize (i.e. give justification as to why a certain decision has been made) the predictions.

9.1. Transparency

Methods that involve *rule extractions* for opening-up non-transparent models have been suggested in the literature. Baesens et al. (2003) evaluated and contrasted three neural network rule extraction techniques, namely, Neurorule, Trepan, and Nefclass for credit-risk evaluation. They concluded that neural network rule extraction is an effective and powerful management tool which allows the development of advanced and user-friendly decision-support systems for credit-risk evaluation. Setiono and Liu (1997) proposed a NeuroLinear method for extracting oblique decision rules from neural networks. The experimental results showed that NeuroLinear is effective in extracting compact and comprehensible rules that have high predictive accuracy from neural networks.

The techniques highlighted above are based on rules, and recently techniques such as *SHAP values*, *Partial Dependence* and *Explainable Boosting Machines* that do not require rules have been applied in domains (other than credit scoring). In their empirical study Caruana et al. (2015) proposed intelligible model that can easily explain its predictions. The study applied generalized additive models with feature interactions (also referred to as explainable boosting machines) on real health care problems. Their results showed that the proposed generalized additive model can perform comparably with best performing machine learning models such as random forest and logistic regression. Lundberg and Lee (2017) used SHAP (SHapley Additive exPlanations) as a unified framework to interpret predictions. The SHAP method assesses contribution of each feature towards a prediction. On the other hand, the partial dependence checks how the prediction changes when different feature values are used. The partial

dependence plots show the marginal impact one or two features have on the prediction of a machine learning model Friedman (2000).

9.2. Deep Learning in Credit Scoring

This section covers deep learning techniques in credit scoring. There is an emerging trend to replace statistical and classical machine learning techniques with deep learning techniques in credit scoring. For instance, Luo et al. (2017) used Corporate Default Swaps (CDS) data to compare performances of deep belief networks with well-known credit scoring models such as logistic regression, multi-layer perceptron and support vector machine. Deep belief networks showed better performance. In their study, Ramasamy and Rajaraman (2017) compared meta-cognitive restricted boltzman machine with extreme learning machines, support vector machines and multi-layer perceptron using Australian, German and Kaggle credit datasets. The meta-cognitive restricted boltzman machine showed superior performance. Tomczak and Zieba (2015) assessed and compared performances of classification restricted boltzman machine with several traditional statistical and machine learning models, such as logistic regression, decision trees, adaboost, random forest etc., using German, Australian, Kaggle and Short-Term Loans data. The classification restricted boltzman machine showed comparable results on German, Kaggle and Short-Term Loan datasets, and better results on Australian dataset. Tran et al. (2016) proposed a hybrid genetic programming and stacked auto-encoder network model. The proposed hybrid model was compared to logistic regression, k -NN, support vector machines, artificial neural networks and decision trees using German and Australian credit datasets. The hybrid model showed a better accuracy rate. Yeh et al. (2015) used daily stock returns to predict defaults and non-defaults using deep belief network and support vector machines. The proposed deep belief network outperformed support vector machines. In their empirical study, Neagoe et al. (2018) compared deep convolutional neural networks with multi-layer perceptron using German and Australian datasets. The results showed a superior overall accuracy rate for deep convolutional neural networks. These studies have shown the superiority of deep learning models in credit scoring. However, Hamori et al. (2018) argue that the performance of deep learning models is dependent on the choice of activation function, the number of hidden layers and the dropout rate. The results in (Hamori et al., 2018) showed a better performance for ensemble methods, such as boosting and bagging, when compared with deep neural networks using Taiwan credit dataset. These studies highlight and reiterate the applicability of deep learning algorithms to credit scoring data.

9.2.1. Data Augmentation

Deep learning models require more data for training to avoid overfitting (Shorten and Khoshgoftaar, 2019). Data augmentation is normally performed to increase the number of training data points. This is done by applying several distortions to the original training images, such as changing the brightness and

rotation of images and the distortions should not change the spatial pattern of target classes (Gómez-Ríos et al., 2018). In Gómez-Ríos et al. (2018), several data augmentations such as *random shift*, *random zoom*, *random horizontal flip* and *random rotation* were applied on image data. The random shift randomly shifts the images by a factor, the random zoom randomly zooms the images by a certain range, the random rotation randomly rotates the images by a certain angle and the random horizontal flip randomly flips the images horizontally to produce additional images. Their results Gómez-Ríos et al. (2018) showed that data augmentation did not significantly improve the accuracy (i.e. accuracy increased by 1%). Kvamme et al. (2018) argue that data augmentation has been mostly done in image processing, and does not necessarily generalize well to other data sources.

On the other hand, (Krizhevsky et al., 2012) achieved significant improvements on error rates when data augmentation was applied on *ImageNet* dataset. A huge deep neural network (60 million parameters and 650,000 neurons) was used for this task, and for this they needed to increase the dataset size. Perez and Wang (2017) used data augmentation on *gold fish vs. dogs* and *cats vs. dogs* datasets. The results showed better improvements in terms of accuracy on both datasets. Salamon and Bello (2016) combined deep learning neural network with data augmentation to classify *audio* data. They used four different data augmentations and the proposed combination significantly outperformed deep neural network without augmentation. Frid-Adar et al. (2018) proposed a combination of classic data augmentation with *Generative Adversarial Network (GAN)* data augmentation. The GAN was used to synthesise new *images of liver lesion*. This combination resulted in significant accuracy improvement. All of these studies used image datasets and credit scoring data is not in the form of images. However, the credit scoring data can be converted into images Zhu et al. (2018).

The above authors have contrasting views on the effectiveness of data augmentation on model accuracy. The conclusion from Krizhevsky et al. (2012) is that when data augmentation is used in conjunction with large deep neural networks, the accuracy increases significantly compared to using data augmentations with "shallow" neural networks. Also Salamon and Bello (2016) found that class accuracy is influenced differently by each augmentation.

10. Limitations in Credit Scoring Literature

Below is the list of limitations which have been partially or completely ignored in credit scoring literature:

- ✓ No inclusion of macro-economic variables;
- ✓ The time it takes for borrowers to default is not determined;
- ✓ Exploratory Data Analysis: detection of outliers and distribution of variables (checking zero-variance for variables) is not performed;
- ✓ Time/Model Complexity is not factored;

- ✓ Creation of new features instead of performing feature space reduction techniques/feature selection is not taken into account;
- ✓ Correlation between dependent variable (or target variable) and independent variables is not assessed.
- ✓ Most studies in literature focus on homogeneous base classifiers and ignore heterogeneous base classifiers for ensemble methods;
- ✓ Using different cut-offs (instead of 0.5) to classify borrowers as either non-defaults or defaults is not covered in literature;
- ✓ Few studies incorporate Type II error. The type II error is more costly in credit scoring, since Type II (False Negative ratio) predicts a borrower as a good borrower but in actual fact he or she is a bad borrower.

The increases in macro-economic variables such as interest rate, inflation rate and unemployment rate may increase the risk of a borrower defaulting. Hence, it is key to incorporate macro-economic variables in credit scoring. Since macro-economic variables are time-varying, it is key to develop forward-looking credit scoring techniques. The survival analysis can cater for forward-looking credit scoring techniques. This can help not only to classify a borrower as default but to know when will a borrower default. The univariate analysis can help in identifying non-predictive variables and also detect outliers. Model/time complexity stems from models with too many parameters. Hence, it is key to keep a few parameters during model development. Although feature reduction/selection improves model accuracy, the literature needs to look at other feature engineering techniques such as taking the sum/product of two features to create a new feature. In cases where there are few independent features, a correlation between the target variable and the independent features can be performed to determine the high predictive features. The literature needs to focus on other ensemble techniques where base classifiers/learners are heterogeneous (i.e. models coming from different model classes). The default cut-off for classifying borrowers is 0.5, e.g. in logistic regression, the literature needs to assess the impact of using different cut-offs. The error which is of interest in credit scoring community is Type II error and the aim is to minimize this error. The literature needs to report more on this error when developing credit scoring models.

11. Results

This section focuses on the meta-analysis of the results from primary studies. The frequency distributions of feature extraction techniques, the models and evaluation metrics from all primary studies are analyzed. The pie-charts for distribution of data balancing techniques and transparency techniques are also shown. The German and Australian credit datasets are selected for model performance comparison, as they are the most frequently used datasets in credit scoring.

11.1. Data Balancing

Credit risk data is generally imbalanced (see Table 3). The non-default borrowers are usually more than the defaulted borrowers. In cases where there

Dataset	Sample Size	#Goods	#Bads	# of features
European Credit Bureau	186,574	179,544	7,030	324
UK	30,000	28,800	1,200	14
Barbados	21,117	20,614	503	20
Indonesia	14,700	14,290	4,410	31
Benelux 2 (Belgium, Netherlands, and Luxembourg)	7,190	5,033	2,157	28
Brazilian	4,504	4,144	360	5
Benelux 1 (Belgium, Netherlands, and Luxembourg)	3,123	1,040	2,083	27
University of California, San Diego	2,435	1,836	599	38
China	1,057	552	505	10
German	1,000	700	300	20
Iranian	1,000	950	50	27
Australian	690	307	383	14
Japanese	653	296	357	15
Polish	240	128	112	30
Texas Banks	162	81	81	19

Table 3: Datasets that are used in literature for credit scoring. Each dataset has a number of good borrowers and a number of bad borrowers.

is a high imbalance in data, the models turn to be biased towards the majority class. To mitigate this bias, techniques such as over-sampling or under-sampling are usually performed. However, this study shows that only 18% of the primary studies in this literature survey have balanced their datasets (see Figure 8). The most used technique is under-sampling the majority class (see Figure 9).

11.2. Feature Extraction

The distribution of feature extraction techniques, such as feature selection and feature engineering for this literature survey is shown in Figure 10. This shows that out of 17 studies (see Table 4 for details) which used feature extraction, Rough Set technique was the most frequently used feature extraction technique, followed by Stepwise, Genetic Algorithm and PCA.

11.3. Evaluation Metrics

Based on this current literature survey, Figure 11 shows commonly used evaluation metrics in credit scoring. The most frequently used evaluation metrics are PCC and AUC (see Table 5 for details).

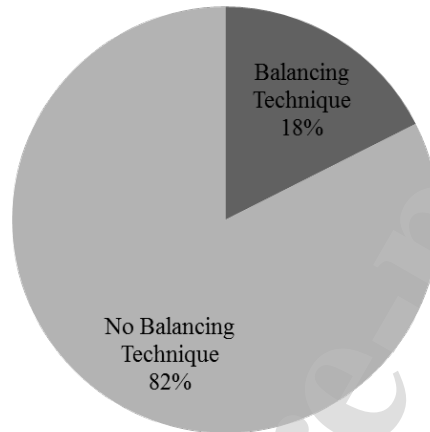


Figure 8: A pie-chart that shows a proportion of literature papers that used data balancing techniques and a proportion of papers that did not balance their datasets.

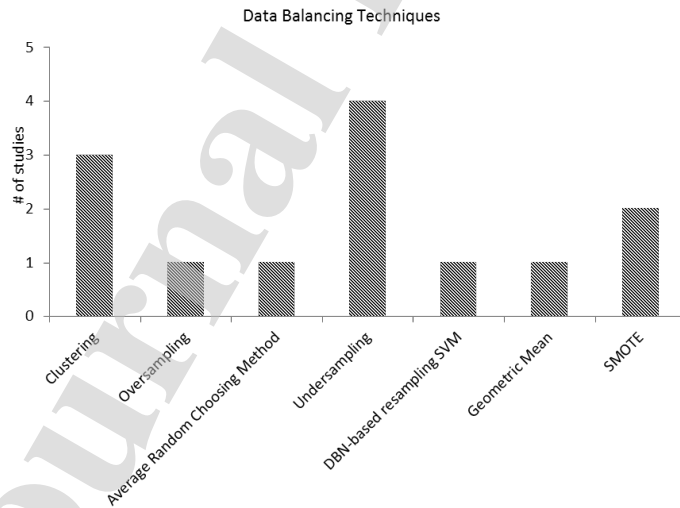


Figure 9: A frequency distribution of data balancing techniques that are used in primary studies of this literature survey.

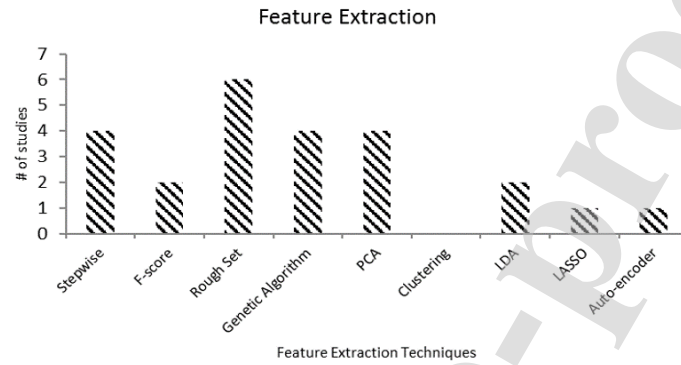


Figure 10: A frequency distribution of feature extraction and engineering techniques that are used in primary studies of this literature survey.

Source	ST	FS	RS	GA	PCA	AE	LDA	LASSO
Chen and Li (2010)		✓	✓					
Kiani and Mahmoudi (2010)	✓						✓	
Wang et al. (2010)			✓					
Zhang et al. (2010)			✓					
Farquad et al. (2011)					✓			
Ping and Yongheng (2011)	✓		✓					
Chi and Hsu (2012)				✓				
Brown and Mues (2012)	✓							
Wang et al. (2012)			✓					
Bijak and Thomas (2012)	✓							
Han et al. (2013)					✓			
Shi et al. (2013)			✓					
Chen and Shi (2013)		✓						
Li et al. (2014)					✓			
Oreski and Oreski (2014)				✓				
Tran et al. (2016)				✓				
Maldonado et al. (2017)								✓
Sutrisno and Halim (2017)								
Mancisidor et al. (2018)					✓	✓		
Zhang et al. (2018)							✓	
Jadhav et al. (2018)				✓				

Table 4: Feature extraction methods used in primary studies. Legend: ST (Stepwise), FS (F-score), RS (Rough Set), GA (Genetic Algorithm), PCA (Principal Component Analysis), AE (AutoEncoder), LDA (Linear Discriminant Analysis), and LASSO (Least Absolute Shrinkage and Selection Operator).

Source	PCC	AUC	T1	T2	F-M	G-M	K-S	SP	SE
Dong et al. (2010)	✓								
Wang et al. (2010)		✓			✓			✓	✓
Twala (2010)	✓								
Hsieh and Hung (2010)	✓								
Kiani and Mahmoudi (2010)	✓								
Chen and Li (2010)		✓							
Wang et al. (2010)		✓			✓			✓	✓
Zhang et al. (2010)	✓								
Yu et al. (2011b)	✓		✓	✓					
Farquard et al. (2011)	✓								
Wang et al. (2011a)	✓		✓	✓					
Wang et al. (2011b)	✓							✓	✓
Ribeiro and Lopes (2011)	✓		✓	✓				✓	✓
Yap et al. (2011)			✓	✓				✓	✓
Yu et al. (2011b)		✓							✓
Ping and Yongheng (2011)	✓								
Louzada et al. (2011)		✓							
Chi and Hsu (2012)		✓					✓		
Marqués et al. (2012a)	✓		✓	✓					
Marqués et al. (2012b)		✓	✓	✓					
Brown and Mues (2012)		✓							
Tang and Qiu (2012)	✓								
Wang et al. (2012)					✓			✓	✓
Bijak and Thomas (2012)		✓					✓		
Louzada et al. (2012)	✓							✓	✓
Shi et al. (2013)	✓								
Chen and Shi (2013)	✓							✓	✓
Han et al. (2013)	✓								
Tsai et al. (2014)									
Abellán and Mantas (2014)		✓							
Oreski and Oreski (2014)	✓								
Li et al. (2014)	✓								
Harris (2015)		✓							
Yi and Zhu (2015)	✓								
Jones et al. (2015)		✓							
Tomczak and Zieba (2015)		✓				✓		✓	✓
Chen and Xu (2015)	✓								
Yeh et al. (2015)	✓								
Zhao et al. (2015)									
Florez-Lopez and Ramon-Jeronimo (2015a)	✓	✓	✓	✓					
Florez-Lopez and Ramon-Jeronimo (2015b)		✓							
Aláraj and Abbod (2015)	✓		✓	✓					
Aláraj and Abbod (2016)		✓							
Tran et al. (2016)	✓		✓	✓					
Yu et al. (2016)	✓		✓	✓					
Xiao et al. (2016)	✓								
Barboza et al. (2017)	✓	✓	✓	✓					
Bequé and Lessmann (2017)	✓	✓							
Lawi et al. (2017)	✓								
Sutrisno and Halim (2017)		✓	✓	✓			✓		
Luo et al. (2017)	✓	✓							
Li et al. (2017a)	✓					✓			
Bequé and Lessmann (2017)	✓		✓	✓					
Chen and Xiang (2017)	✓								✓
Li et al. (2017b)	✓							✓	✓
Ramasamy and Rajaraman (2017)						✓			✓
Okesola et al. (2017)	✓								
Chen et al. (2017)	✓		✓	✓					
Maldonado et al. (2017)		✓							
Abellán and Castellano (2017)	✓	✓							
Jadhav et al. (2018)	✓	✓							
Vanderheyden and Priestley (2018)							✓		
Kvamme et al. (2018)	✓	✓						✓	✓
Mancisidor et al. (2018)		✓					✓		
Zhu et al. (2018)	✓	✓					✓		
Martey Addo et al. (2018)		✓							
Neagoe et al. (2018)	✓								
Hamori et al. (2018)	✓	✓			✓				
Zhang et al. (2018)	✓								
Xia et al. (2018)	✓								
Yu et al. (2018)	34 ✓	✓							
Chang et al. (2018)		✓							
Li et al. (2018)		✓							

Table 5: Evaluation metrics from primary studies. Legend: PCC (Percentage Correctly Classified), AUC (Area Under ROC Curve), T1 (Type I Error), T2 (Type II Error), F-M (F-measure), G-M (G-measure), K-S (Kolmogorov-Smirnov Statistic), SP (Specificity), SE (Sensitivity)

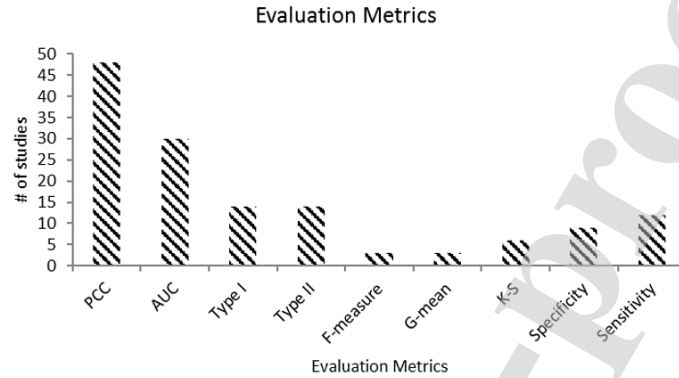


Figure 11: A frequency distribution of evaluation metrics that are used in primary studies of this literature survey.

11.4. Models

This literature survey focuses on statistical, traditional and state-of-the-art machine learning models. Based on the results of this literature survey, Figure 12 shows that LR, SVM and ANN were the most frequently used single classifiers (see Table 6 for details). For ensemble of classifiers, Boosting was the most frequently used ensemble classifier. The deep learning classifiers are among the least frequently used classifiers and this is attributable to the fact that deep learning classifiers are not applied extensively in credit scoring.

11.5. Transparency

Under Basel II Accord (Basel Committee on Banking Supervision, 2006), credit scoring models are required to be transparent (i.e. to explain their predictions). The majority of machine learning models are not transparent in nature. Hence, a need to make the models transparent in credit scoring is paramount. However, as can be seen in Figure 13, this survey shows that only 8% of the primary studies have looked into techniques which make models transparent.

11.6. Model Performance on German and Australian Datasets

Out of 74 primary studies, 39 primary studies either used German credit dataset, Australian credit dataset or both. The results for both PCC and AUC on both datasets were collected from the primary studies. The results (see Figure 14) show that on average on both datasets, the Convolutional Neural Networks performed better in terms of PCC, followed by ensemble classifiers such as Random Forests, Bagging and Boosting. The high performance of CNN can be attributed to the fact that they can detect features that are more discriminative between borrowers. In terms of AUC, the ensemble of classifiers

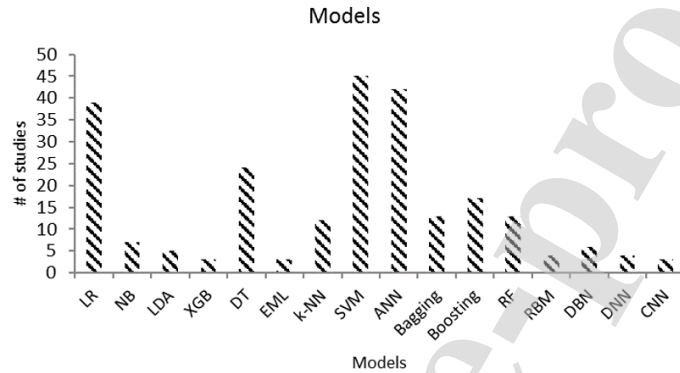


Figure 12: A frequency distribution plot of most frequently used statistical and machine learning models in credit scoring.

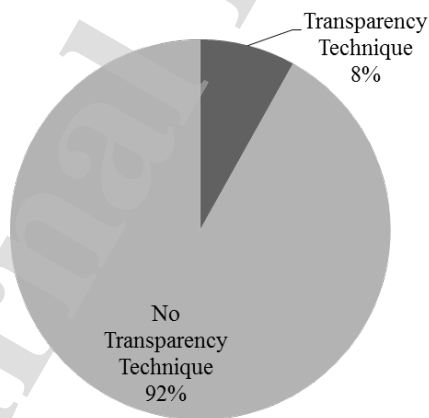


Figure 13: A pie-chart that shows a proportion of studies that used transparency techniques and a proportion of studies that did not use transparency techniques.

Source	LR	NB	LDA	XGB	DT	EML	k-NN	SVM	ANN	BA	BO	RF	RBM	DBN	DMLP	CNN
Dong et al. (2010)	✓								✓							
Kiani and Mahmoudi (2010)					✓					✓						
Zhang et al. (2010)	✓	✓			✓		✓			✓						
Twala (2010)									✓	✓						
Hsieh and Hung (2010)								✓		✓						
Wang et al. (2010)	✓							✓		✓						
Chen and Li (2010)								✓		✓						
Yu et al. (2011b)								✓		✓						
Farquad et al. (2011)								✓		✓						
Wang et al. (2011a)	✓				✓				✓	✓						
Wang et al. (2011b)									✓	✓						
Ribeiro and Lopes (2011)													✓		✓	
Louzada et al. (2011)	✓															
Yu et al. (2011b)	✓	✓	✓				✓									
Ping and Yongheng (2011)								✓								
Yap et al. (2011)	✓				✓											
Chi and Hsu (2012)	✓															
Marqués et al. (2012a)	✓	✓			✓		✓		✓	✓						
Brown and Mues (2012)	✓		✓		✓				✓		✓	✓				
Marqués et al. (2012b)	✓				✓		✓		✓		✓					
Bijak and Thomas (2012)	✓				✓					✓	✓					
Louzada et al. (2012)	✓															
Tang and Qiu (2012)								✓								
Wang et al. (2012)	✓				✓				✓							
Han et al. (2013)	✓								✓							
Chen and Shi (2013)									✓							
Shi et al. (2013)									✓							
Chen and Shi (2013)									✓							
Tsai et al. (2014)						✓			✓	✓						
Abellán and Mantas (2014)						✓										
Li et al. (2014)								✓					✓			
Oreski and Oreski (2014)									✓							
Harris (2015)	✓								✓							
Yi and Zhu (2015)									✓							
Jones et al. (2015)	✓		✓						✓		✓	✓				
Zhao et al. (2015)									✓							
Florez-Lopez and Ramon-Jeronimo (2015b)	✓		✓		✓		✓		✓		✓	✓				
Chen and Xu (2015)									✓							
Yeh et al. (2015)	✓								✓							
Tomczak and Zieba (2015)	✓				✓				✓		✓	✓	✓			
Aláraj and Abbod (2015)	✓								✓		✓	✓	✓			
Florez-Lopez and Ramon-Jeronimo (2015a)	✓		✓		✓		✓		✓		✓	✓				
Xiao et al. (2016)										✓						
Tran et al. (2016)	✓				✓		✓		✓		✓				✓	
Yu et al. (2016)	✓				✓				✓					✓		
Aláraj and Abbod (2016)	✓				✓				✓		✓					
Abellán and Castellano (2017)	✓				✓				✓		✓					
Chen et al. (2017)	✓				✓				✓		✓					
Barboza et al. (2017)	✓								✓		✓	✓				
Bequé and Lessmann (2017)	✓				✓		✓		✓		✓					
Lawi et al. (2017)	✓								✓		✓					
Okesola et al. (2017)		✓														
Maldonado et al. (2017)								✓								
Sutrisno and Halim (2017)	✓								✓							
Luo et al. (2017)	✓								✓					✓		
Li et al. (2017a)	✓								✓						✓	
Bequé and Lessmann (2017)	✓				✓	✓	✓		✓							
Chen and Xiang (2017)	✓								✓							
Li et al. (2017b)	✓								✓					✓		
Ramasamy and Rajaraman (2017)						✓			✓				✓			
Zhu et al. (2018)	✓								✓			✓				✓
Martey Addo et al. (2018)	✓								✓			✓			✓	
Neagoe et al. (2018)									✓							✓
Zhang et al. (2018)		✓			✓		✓		✓			✓				
Xia et al. (2018)	✓			✓	✓			✓				✓				
Vanderheyden and Priestley (2018)	✓															
Kvamme et al. (2018)																✓
Mancisidor et al. (2018)																
Chang et al. (2018)				✓												
Li et al. (2018)				✓												
Anqi Cao (2018)	✓		✓		✓				✓			✓				
Jadhav et al. (2018)		✓					✓		✓					✓		
Yu et al. (2018)								✓								
Hamori et al. (2018)									✓	✓	✓	✓			✓	

Table 6: Models used in credit scoring. Legend: LR (Logistic Regression), NB (Naïve Bayes), LDA (Linear Discriminant Analysis), XGB (XGBoost), EML (Extreme Learning Machines), k-NN (*k*-Nearest Neighbour), SVM (Support Vector Machine), ANN (Artificial Neural Network), BA (Bagging), BO (Boosting), RF (Random Forest), RBM (Restricted Boltzman Machine), DBN (Deep Belief Network), DMLP (Deep Multi-Layer Perceptron), and CNN (Convolutional Neural Network).

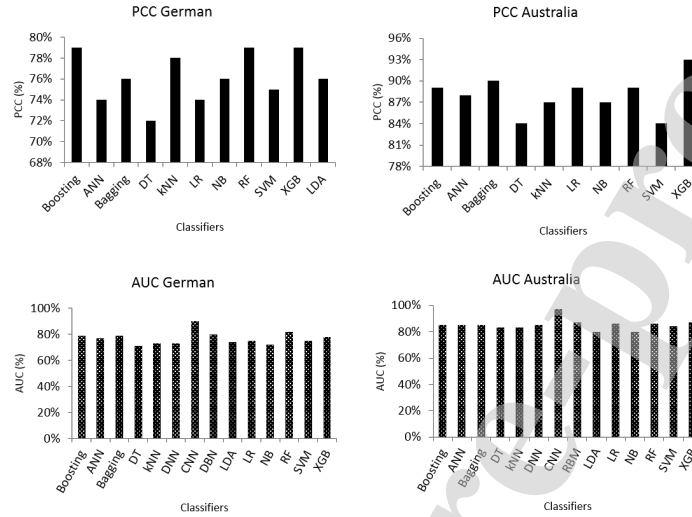


Figure 14: Averages of pcc and auc that are calculated from the results reported on the papers considered for this literature survey that used German and Australian credit datasets.

performed better than all single classifiers. We noted that different studies reported differently on PCC and AUC for each of the classifiers. This could be attributed to data pre-processing of each study.

12. Guiding Machine Learning Framework for Credit Scoring

Based on the results of this literature survey, we propose a framework which is shown in Figure 15, which serves as a guideline for credit scoring analysts. The nature of the data in credit scoring is either balanced or imbalanced. For both types of datasets, this framework suggests exploratory data analysis, data pre-processing, feature extraction techniques, the sampling methodology, the models to use as benchmarks, the best performing models in credit scoring, the evaluation metrics and the technique which explains model predictions. This literature survey showed that the most frequently used techniques for feature extraction are Rough Set and Genetic Algorithm. The sampling methodology that we are suggesting for imbalanced datasets is SMOTE. The sampling is only applied on the training set. The models which can be used as benchmarks is Logistic Regression (LR) and Decision Trees (DT). The LR performs similarly to most traditional Machine learning models and the choice of selecting DT is its capability of explaining predictions. This literature survey shows that ensemble classifiers and CNN performed better when compared to other models, hence we suggest these two models for use in credit scoring. For imbalanced datasets,

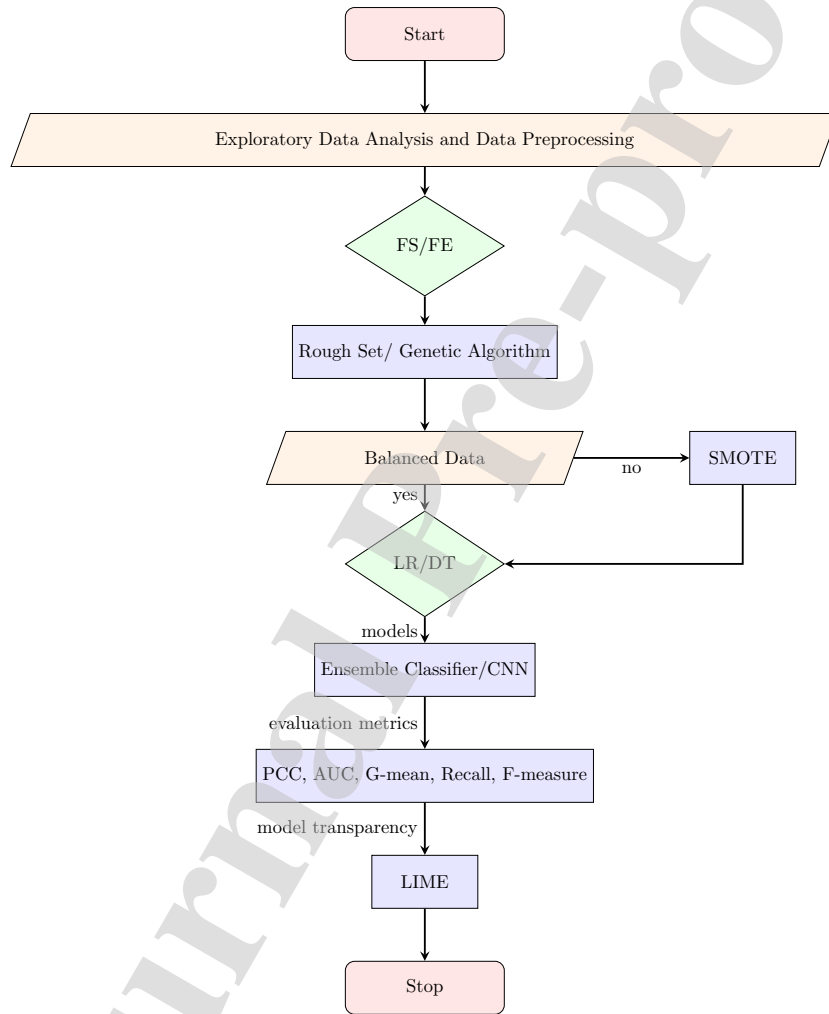


Figure 15: The guiding machine learning credit scoring framework that is proposed in this literature survey.

evaluation metrics which can be applied are G-mean, Recall and F-measure. Since predictions from ensemble classifiers and CNN cannot be explained, we suggest LIME for predictions' explanations.

13. Conclusion and Future Work

Many studies over the years have evaluated and contrasted the performances of different statistical and classical machine learning models in credit scoring. However, no consensus has been reached in identifying the best performing model. This literature survey systematically reviewed the most commonly used statistical, classical machine learning and deep learning models in credit scoring. The performances (on German and Australian credit datasets) of statistical, classical machine learning and deep learning models that were reported in literature were compared in this literature survey. The literature results showed that an ensemble of classifiers generally outperform single classifiers. Despite the minimal application of deep learning models in credit scoring literature, deep learning models such as convolutional neural networks showed better results compared to statistical and classical machine learning models. This literature survey also highlighted limitations in credit scoring literature. The credit scoring literature often ignores exploratory data analysis, omits inclusion of macro-economic variables and does not determine correlation between the target variable and the independent features, to mention a few. Furthermore, in this survey we proposed a guiding machine learning framework for analysts to perform credit scoring. This framework includes feature selection methods, data balancing technique, models to consider for bench-marking, best performing models in credit scoring, relevant evaluation metrics and transparency method. This survey pointed to emerging directions in credit scoring such as the use of techniques that make model predictions explainable and the applications of deep learning models in credit scoring.

Future research should focus more on balancing classes of datasets in credit scoring. Balancing techniques that neither do over-sampling nor under-sampling such as linear dependence approach and wavelet data transformation should be explored in credit scoring. Future research should incorporate macro-economic variables such as interest rates, unemployment rates and inflation rates. An increase in any of the mentioned macro-economic variables may increase the risk of a borrower defaulting. Future studies should consider the time it takes for borrowers to default and this will allow commercial banks to be forward-looking and proactive. The literature does not take into consideration the exploratory data analysis, hence future research should focus on this aspect to better understand for example the distributions of features. Future studies should focus more on time/model complexity to allow efficiency in model development in credit scoring. Future research should determine the correlation between the target variable and the independent variables/features in-order to identify predictive variables/features. There are few studies that focused on using heterogeneous base classifiers for ensemble methods (Li et al., 2018; Xia et al., 2018; Hsieh and Hung, 2010). However, future studies should focus more on using heterogeneous

instead of homogeneous base classifiers in ensemble methods to allow diversity of base classifiers.

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Ensemble classifiers generally outperform single classifiers in credit scoring. Deep Learning techniques have yet to be used in credit scoring to replace traditional machine learning methods.

Deep learning architectures such as Convolutional Neural Networks outperform statistical and traditional machine learning models.

The non-transparent nature of Deep Learning methods is the main bottleneck for them to replace traditional interpretable machine learning methods in credit scoring.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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